

Presubmission Document

Domain: Logistics and Supply chain

Client: Global Freight Solutions (GFS)

1. Business Problem Understanding and Objective Formulation

Global Freight Solutions (GFS) is a rapidly growing logistics company operating across multiple cities, suppliers, warehouses and transportation routes. Over the past 18 months, the company has experienced increasing operational costs, shipment delays, declining customer satisfaction, inconsistent supplier performance, warehouse capacity constraints and rising transportation and penalty costs.

Currently, GFS relies on reactive decision-making instead of proactive measures, making it difficult to identify the root causes of these operational inefficiencies. A supplier delay or warehouse bottleneck causes a shipment delay, which triggers a customer complaint, which results in a penalty cost, ultimately impacting Finance.

The objective is to analyse historical shipment data, identify key drivers of operational failures and develop predictive models to enable proactive decision-making across the supply chain.

Potential Business Problems from Data Set

DELAY RATE	TOTAL PENALTIES	COMPLAINT RATE	AVG CUSTOMER SATISFACTION	AT-RISK SHIPMENTS
15.3% 7,726 of 50,500 shipments	₹4.1Cr 100% from delayed shipments	11.1% Spikes 44.7% on delays	3.69 Out of 5.0 below target	40.4% Carry ≥1 avoidable risk factor

Delay Drivers

WEATHER	WAREHOUSE CAPACITY	SUPPLIER PERFORMANCE	ROUTE SELECTION	FLEET MAINTENANCE
65.2% Storm warnings create catastrophic 65.2% delay rates	31.8% Warehouses above 85% utilisation drive a 3× delay rate	34.8% Repeat-offender suppliers are severely under penalised	22.5% High-risk routes (score 67–99) carry double the delay rate	17.1% Overdue maintenance (8–30 days) adds measurable delay risk

Business Objectives	Analytical Objectives
Reduce shipment delay rate and enable early intervention	Binary classification on delay_flag
Control and forecast transportation costs for route planning	Regression on transportation_cost
Quantify financial risk from penalty exposure on delayed shipments	Two-stage model (deterministic split + Regression) on penalty_cost
Improve Customer Satisfaction	Reducing delivery delays is expected to improve customer satisfaction from 2.0 to 4.0 on average so separate modelling of customer satisfaction was not required.
Identify the inconsistent suppliers and improve the supply chain performance	Supplier clustering to assign risk tiers based on delay history and performance.

Warehouse bottlenecks are root cause findings from exploratory data analysis, not separate modelling objectives - it feeds into the classification model as features.

2. Challenges and Constraints

2.1 Data Challenges:

- Class Imbalance of 5.5:1 (84.8% on time vs 15.2% delayed). Naive models achieve 84.8% accuracy without learning anything useful. F1 and AUC-ROC are used as evaluation metrics.
- Feature selection was applied to remove irrelevant columns and high-cardinality identifiers (supplier_internal_code 48K unique values) from the 39 raw features to avoid overfitting.
- Three features (supplier_rating, vehicle_age_years and rainfall_mm) contain missing values and showed near zero correlation with their respective target variables so excluded from modelling.
- Penalty cost magnitude lacks key predictive features like delay duration, SLA breach tier etc.
- supplier_region contains inconsistent values (N/North,S/South) formats so it required standardisation.
- June 2026 data was incomplete and excluded from trend analysis.

2.2 Assumptions/Limitations/Constraints:

- The 80:20 train test split is sufficient for generalisation of 50,000 records.
- The model treats each shipment as an independent event, ignoring any sequential dependencies between shipments on the same route or from the same supplier on the same day.
- Heteroskedasticity and leptokurtic residuals observed in cost distributions violate linear regression assumptions which means the linear regression model is less reliable for expensive routes and more reliable for normal routes.
- The transportation cost series exhibits steady-state behavior with an overall mean of ₹17,554 and a monthly variation of approximately 8%. The temporal stability assumption for regression analysis is considered satisfied.
- Model performance on unseen routes, new suppliers/warehouses is unknown as they are not part of training data.

3. Background Study / Literature Review

- A Machine Learning-Based Framework for Predicting Supply Delay Risk Using Big Data Technology, International Journal of Data Science and Analytics, Springer Nature, December 2025, <https://link.springer.com/article/10.1007/s41060-025-00969-8>
- Enhancing Supply Chain Resilience: ML Approach for Predicting Product Availability Under Disruption, ScienceDirect, January 2024, <https://www.sciencedirect.com/science/article/abs/pii/S0957417424000915>
- Deep Learning Framework for Interpretable Supply Chain Forecasting Using SOM, ANN and SHAP, Scientific Reports, Nature, July 2025, <https://www.nature.com/articles/s41598-025-11510-z>

4. Proposed Methodology / Solution Approach

4.1 Preprocessing:

- Duplicate entry records were removed.
- Train/test split (80/20) performed before any encoding to prevent data leakage.
- Categorical variables were encoded using one-hot encoding (drop_first=False) and numerical features were scaled using StandardScaler to ensure consistent feature magnitude across models.
- Multicollinearity and correlation analysis were performed to identify and drop redundant and near-zero correlation features.

4.2 Modelling Tracks:

- Delay Classification: Three models will be benchmarked (Logistic Regression, Random Forest and XG Boost) on identical train/test splits & selecting production model based on AUC, cross-validation stability and feature interpretability.

- **Transportation Cost Regression:** Linear Regression was evaluated as the primary model given the strong linear relationship between route_distance_km and transportation cost ($R^2 = 0.88$). Gradient Boosting Regressor was tested as a comparison model but Linear Regression outperformed (Log transformation of the target variable may be explored in future iterations to address the observed heteroskedasticity in residuals).
- **Penalty Cost - Stage 1:** Deterministic split using delay_flag to separate zero vs. non-zero penalty rows, no classifier required. **Stage 2:** Linear Regression and Gradient Boosting Regressor evaluated on delayed subset (7,652 rows). Both returned $R^2 \sim 0$ indicating no predictive signal in the available features.

4.3 How the Solution Addresses the Problem:

The classification model on delay_flag enables GFS to identify high risk shipments before dispatch, allowing operations teams to intervene proactively. The transportation cost regression supports route planning and budget forecasting by quantifying the cost drivers. The two stage penalty cost model aims to identify which shipments carry penalty risk and quantifies financial exposure on delayed shipments. Supplier clustering identifies high risk suppliers, enabling procurement teams to take targeted corrective actions. Together, these ML models shift GFS from reactive manual reporting to proactive, data-driven decision-making.

5. Proposed Timeline

Phase	Activity	Duration
Data Understanding	Problem definition, data exploration	Week 1
Data Preprocessing	Cleaning, imputation/encoding, standardization, feature engineering	Week 2
Exploratory Data Analysis (EDA)	Trend analysis, visualizations, correlation analysis	Week 3
Model Development	Baseline models, ensemble models, cross-validation, imbalance handling	Week 4-5
Evaluation	Residual diagnostics, heteroskedasticity tests, performance metrics, model selection	Week 6-7
Results Interpretation	Conclusion, Technical documentation, Dashboard(Visualization), business recommendations	Week 8-10
Presentation Prep	Final presentation, report and deployment proposal	Week 11-12

6. Expected Deliverables / Outputs

- **Data Analysis Report** containing exploratory data analysis, feature selection and business insights.
- **Delay Prediction Model** for identifying shipments at risk of delivery delays.
- **Penalty/Transportation Cost Prediction Model** for estimating potential financial penalties/transportation cost.
- **Performance Evaluation Report** including performance assessment, cross-validation and comparison of alternative modelling approaches.

- **Data Visualizations and Dashboard** for operational insights and KPI monitoring.
- **Deployment Proposal:** Recommendations for integrating the delay classification model into GFS operational systems for real-time shipment risk scoring.
- **Technical Documentation** covering data preprocessing, feature engineering, model development and evaluation.
- **Final Presentation Report** summarizing methodology, findings, recommendations and expected business impact.

6.1 Projected Business Impact:

Note: *The following projections are estimated based on EDA findings and model outputs. Actual impact will depend on operational implementation.*

- Projected 44.7% reduction in delay-related complaints.
- Projected reduction in delay rate from 31.8% to 12% through warehouse capacity triggers.
- Estimated ~₹0.87 Cr annual savings through supplier performance management.
- The historical penalty exposure of approximately ₹4.1 Cr indicates the potential financial benefit of reducing shipment delays.
- Estimated ~₹1.8 crore annual savings through storm-aware dispatch planning.
- Projected improvement in premium customer satisfaction by approximately 22.5%.